Jin Tao

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Education

Technische Universität Berlin — Berlin, Germany
M.Sc. in Computational Engineering Science; GPA: 1.3 (1.0 best) — *Feb. 2025 – Present*

- Thesis and one 6-credit course remaining; to be completed in personal time alongside full-time work.

Technische Universität Berlin — Berlin, Germany
M.Sc. in Vehicle Engineering — *Sep. 2020 – Feb. 2025*

Hainan University — Hainan, China
B.Sc. in Vehicle Engineering — *Sep. 2015 – June 2019*

Experience

Researcher (full-time) — Aug. 2026 – Present
Working Student — *May 2026 – July 2026*
Center for Tangible Artificial Intelligence and Digitalization (ZEKI) — *Berlin, Germany*

- Lead development of a TurtleBot4 robotics platform integrating SLAM, navigation, autonomous exploration, semantic mapping, and an AI agent that translates user requests into robot actions.
- Implement a distributed onboard architecture, running SLAM and navigation on Raspberry Pi and semantic mapping and the agent system on NVIDIA Jetson Nano.
- Develop a multi-robot collaboration system for disaster response, integrating UGVs and UAVs for collaborative mapping and task coordination through a central device.

Research Assistant — Oct. 2025 – Apr. 2026
Robotics and Biology Laboratory (RBO) — *Berlin, Germany*

- Designed and built an acoustic gripper from scratch, using acoustic signatures to identify grasped objects.
- Improved the software pipeline for a Franka manipulator by resolving a persistent impedance-controller bug that caused abnormal oscillations at higher impedance settings.
- Restructured the Franka ROS 2 stack and streamlined Git workflows using submodules, establishing clear package responsibilities and boundaries across drivers, controllers, and robot descriptions to improve maintainability and development efficiency.

Working Student — Sep. 2024 – Oct. 2025
Mbition GmbH — *Berlin, Germany*

- Developed a full-stack software system for monitoring test vehicles and managing vehicle bookings.

Software Engineer — Oct. 2022 – Mar. 2024
TU Berlin Formula Student Team - FaSTTUBe — *Berlin, Germany*

- Developed a ROS 2 trajectory-tracking and control package using local and global planned trajectories, with pure pursuit and model predictive control (MPC), targeting minimum lap time.
- Developed a steering control system using cascaded current, speed, and position PID loops to achieve fast, stable tracking of commanded steering positions.

Team Leader — May 2018 – Nov. 2019
Hainan University Formula Student Team — *Hainan, China*

- Designed and built the vehicle's double-wishbone suspension, applying vehicle dynamics principles to determine its geometry.
- Led team organization and coordination, and competed as the endurance-event driver at Formula Student China 2019.

Achievements and Projects

TaoRobot: Autonomous Driving Full Stack | Code & Demo — June 2026 – Present

- Sole developer of an open-source ROS 2 autonomy stack, owning architecture, algorithm implementation, and Gazebo simulation integration.
- Implemented pose-graph SLAM, particle-filter localization, A* path planning with path smoothing, and pure pursuit trajectory tracking.

Magnetic Joint-State Estimation Mar. 2026 – Sep. 2026

- Proposed and developed a ball-joint orientation estimation system using two Adafruit magnetic sensors costing less than €7 each and a magnet to estimate yaw, pitch, and roll.
- Evaluated orientation accuracy against an Xsens MTi-630 IMU, achieving comparable accuracy under the tested conditions without long-term drift.
- Preparing a conference paper with the project team.

Machine Learning for Visuo-Tactile Sensor Fusion Mar. 2026 – Sep. 2026

- Developed training and evaluation infrastructure for estimating in-hand object pose from RGB images and tactile measurements, using ground-truth pose annotations.
- Implemented and compared ten existing methods, then developed a transformer-based fusion model that matched the best evaluated baseline's pose-estimation accuracy.

Master's Thesis: Autonomous Shuttle Pedestrian Following Aug. 2024 – Jan. 2025

- Developed and deployed a pedestrian-following system on a real autonomous shuttle, integrating A* path planning, pure pursuit trajectory tracking, and cascaded PID control.
- Integrated YOLO-based gesture detection to stop, resume driving, and adjust following distance.

GPU-Accelerated Visual SLAM and Navigation Integration Mar. 2025 – Sep. 2025

- Deployed GPU-accelerated visual SLAM on a Jetson Xavier-based mobile robot and integrated its outputs with Nav2 by projecting 3D map points into a 2D occupancy grid.
- Enabled GPU acceleration for visual SLAM on NVIDIA Jetson Xavier, reducing CPU utilization by 40% compared with the original CPU-only pipeline.

Skills

Programming: Python, C/C++, MATLAB/Simulink
Robotics and Simulation: ROS, ROS 2, Nav2, Gazebo
Computer Vision and Machine Learning: YOLO, DINOv3, PyTorch, TensorFlow, TensorRT
Embedded Systems and Hardware: Raspberry Pi, NVIDIA Jetson Nano/Xavier, Arduino, ESP32; soldering, PCB design and debugging
Development Tools: Git, Docker
Agentic Development: Claude Code, Codex; orchestrating coding agents through Git-based workflows, including issues, branches, and pull requests
AI Tools: LangChain, LangGraph, Ollama, llama.cpp
Languages: English (advanced), German (intermediate), Chinese (native)

Selected Coursework (Grades 1.0–1.3; 1.0 Best)

Robotics: 1.0 | **Applications of Robotics and Autonomous Systems**: 1.0
Sensor Systems: 1.3 | **Machine Learning 1**: 1.3 | **Motion Planning**: 1.3
Numerical Analysis I for the Engineering Sciences: 1.0